

# Quantum Annealing for Inverse Kinematics in Robotics

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**Abstract**—This paper explores the application of quantum annealing to solve the inverse kinematics (IK) problem in robotics by reformulating it as a Quadratic Unconstrained Binary Optimization (QUBO) problem. Using a linear binary approximation of trigonometric functions, we transform the IK problem into a form suitable for quantum optimization, demonstrating the potential of quantum annealing in addressing NP-Hard optimization challenges in robotics. We show that Global Embedding on Zephyr is the most efficient strategy, reducing qubit usage and improving QPU access times. For large-scale QUBO problems, we show that hybrid quantum-classical approaches achieve over a 30-fold speedup in optimization tasks. Specifically, for a QUBO instance of size  $900 \times 900$  (corresponding to  $2^{900}$  possible solutions), quantum annealing demonstrates a time-to-solution (TTS) improvement of more than 30 times compared to classical methods. This work underscores the potential of quantum computing and hybrid optimization methods to address traditionally challenging robotics problems like IK, offering enhanced scalability and performance.

## I. INTRODUCTION

Inverse Kinematics (IK) can be viewed as a problem of translating operational space coordinates to the joint-space coordinates for mechanical linkages. In Robotics, IK is a part of the motion planning problem for robot arms or for walking robots. In some cases, there exists an analytical solution to the IK problem; in other cases, it has to be solved numerically.

IK is a non-convex problem. When a good initial guess is available, IK can be solved using local methods: gradient descent [1], sequential quadratic programming [2], convex relaxation [3], etc. Otherwise, the use of global optimization methods might be required. In [4] IK was cast as a mixed-integer convex program (MICP), solving the local minima problem and providing a possibility for integrating IK with other control-related optimizations. However, the drawback to this approach is the lack of rapid convergence guarantees for MICP: in the worst-case scenario, MICP has a computational cost of an exhaustive search, unless terminated. Additionally, the time to solve MICP grows rapidly with the number of integer variables, limiting its viability to problems of moderate size, or to offline computations. This motivates

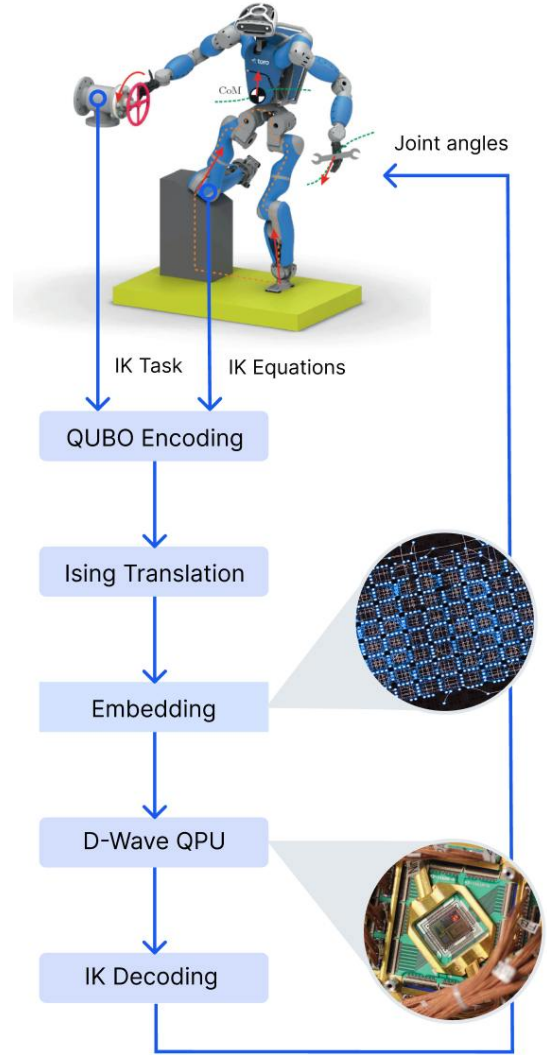


Fig. 1: Workflow for solving inverse kinematics using quantum annealing: Translating joint angle equations into QUBO/Ising models, embedding them on a D-Wave quantum processor, and decoding the quantum solution into actionable results.

the use of quantum annealer solvers as an alternative to integer optimization for the IK tasks.

In the robotics field, many problems exhibit NP-Hard characteristics. NP-Hard problems are those that are at least as difficult as the hardest problems in NP, indicating there is no known polynomial-time algorithm to solve them unless  $P = NP$ . These characteristics often necessitate approxi-

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mation or heuristic strategies, as exact solutions may be computationally prohibitive [5], [6], [7], [8]. Developing an algorithm that employs classical physics properties (Classical Computing) to achieve global solutions in all cases remains an area of ongoing exploration [9], [10], [11]. This limitation stems from the absence of definitive evidence that  $NP = P$ . If  $NP \neq P$ , as widely conjectured, NP-Hard problems cannot be solved in polynomial time by classical means.

Quantum annealing (QA), a specific quantum computing approach implemented in devices like D-Wave's quantum annealers, leverages quantum fluctuations to escape local minima and converge toward global optima. This makes quantum annealing particularly promising for addressing NP-Hard problems.

In this work, we employ quantum annealing to solve a class of IK problems by using the Linear Binary Approximation of trigonometric functions to transform the IK problem into a Quadratic Unconstrained Binary Optimization (QUBO) form, which will be embedded into the quantum annealer, as shown in Figure 1. We thus aim to provide insight into the use of quantum computing for solving robotics problems that share a similar mathematical structure. Therefore, this work serves as a resource for researchers in physics and quantum computing who are interested in exploring such applications, aligning with the scope of this paper.

#### A. Motivation

Quantum annealing can be seen as a new tool for solving discrete, combinatorial optimization problems, particularly those formulated as binary optimization tasks. Theoretical analysis shows that quantum annealing outperforms simulated annealing in convergence to the optimal solution [12]. This advantage arises from quantum tunneling, as shown in Figure 2, which enables quantum annealing to overcome energy barriers that hinder classical methods. Furthermore, quantum annealing has been shown to outperform simulated annealing in practice for many complex problems, reinforcing its value in solving real-world challenges [13], [14].

Although current quantum hardware remains limited in terms of qubit count, coherence times, and noise levels, quantum annealers have shown promising advantages for optimization tasks. They provide a bridge between purely classical techniques and the future promise of fault-tolerant universal quantum computers. As quantum hardware continues to advance, its application to NP-Hard problems, including IK, is likely to expand, offering scalable and efficient solutions that were previously unattainable.

#### B. Novelty

The primary novelty of this work lies in its position as, to the best of the authors' knowledge, the first practical quantum application of IK using real quantum hardware. This represents a new way in addressing the IK problem through quantum annealing technique. Traditionally solved using classical optimization methods that rely on deterministic and sequential approaches, the IK problem is particularly

well-suited to quantum annealing due to its inherent ability to explore multiple solution spaces simultaneously. By leveraging this capability, quantum annealing offers a significant advantage in overcoming local minima and converging to global optima.

#### C. Contributions

The contributions of this research are twofold. Firstly, it introduces a robust framework for addressing the IK problem with quantum computing, establishing a new paradigm for solving complex optimization problems. This contribution not only enriches the field of quantum optimization but also underscores the practical applicability of quantum annealing in real-world engineering scenarios. Secondly, the study provides valuable insights into the capabilities and limitations of current quantum annealing hardware, including D-Wave's quantum processors. While quantum computing remains in its infancy, this research showcases its potential to address NP-Hard problems that have been intractable for classical computing approaches. This work pioneers the use of quantum annealing on real quantum hardware for the IK problem, advancing quantum optimization research and showcasing its potential in robotics and control applications.

#### D. Paper Structure

The paper is organized as follows: Section II provides an overview of Quantum Annealing in D-Wave quantum processing unit (QPU), discussing the principles and mechanisms underlying the quantum annealing process and its implementation on D-Wave's quantum processing unit. Section III introduces QUBO, formulating the IK problem within the framework of Quantum Unconstrained Binary Optimization. Section IV presents Empirical Results and Analysis for Solving IK Using Quantum Annealing, detailing experimental results to evaluate the effectiveness of the quantum annealing approach. Finally, Section V concludes the paper with a summary of key contributions and an outlook in Conclusion.

## II. QUANTUM ANNEALING IN D-WAVE QPU

The standard annealing protocol in the D-Wave QPU involves tuning the quantum system's parameters to solve optimization problems efficiently. This process is governed by the evolution of the Hamiltonian and the dynamic interaction between physical quantities such as magnetic flux and coupling strengths. The D-Wave QPU is a network of superconducting flux qubits that are tunably coupled [15], [16], [17], [18], [19]. The physical Hamiltonian for the flux qubits in the qubit approximation is given by:

$$H = -\frac{1}{2} \sum_i \left[ \Delta_q(\Phi_C(s)) \sigma_x^{(i)} - 2h_i |I_p(\Phi_C(s))| \Phi_{xi}(s) \sigma_z^{(i)} \right] + \sum_{i>j} J_{ij} M_{AFM} [I_p(\Phi_C(s))]^2 \sigma_z^{(i)} \sigma_z^{(j)}, \quad (1)$$

where:

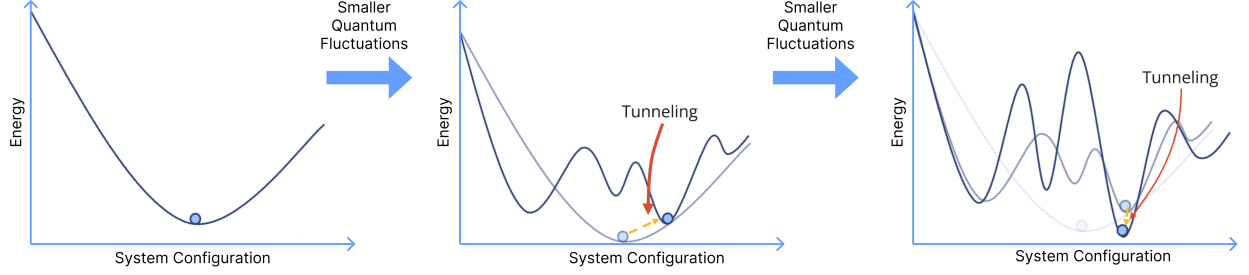


Fig. 2: Schematic representation of the adiabatic quantum annealing process, illustrating how a system transitions from an initial, relatively simple potential landscape to a more complex one while quantum fluctuations are gradually reduced. The red arrows denote quantum tunneling events, through which the system can escape local minima and ultimately converge on the global minimum energy configuration.

- $\Delta_q(\Phi_C(s))$ : Energy difference between the two eigenstates of a qubit at the degeneracy point, determined by the flux  $\Phi_C(s)$  applied to the qubit's Compound-Compound Josephson-Junction (CCJJ) structure.
- $h_i$ : Local field term representing the external flux bias for qubit  $i$ .
- $I_p(\Phi_C(s))$ : Persistent current in the body of the qubit loop, dependent on  $\Phi_C(s)$ .
- $\Phi_{xi}(s)$ : External flux applied to qubit  $i$ .
- $M_{AFM}$ : Maximum mutual inductance generated by the couplers between qubits (typically 2 pH).
- $J_{ij}$ : Coupling strength between qubits  $i$  and  $j$ .
- $\sigma_x^{(i)}, \sigma_z^{(i)}$ : Pauli matrices operating on qubit  $i$  in the  $x$  and  $z$  directions, respectively.

To control the annealing process, the normalized annealing bias  $c(s)$  is introduced:

$$c(s) = \frac{\Phi_C(s) - \Phi_{\text{initial,C}}}{\Phi_{\text{final,C}} - \Phi_{\text{initial,C}}}, \quad (2)$$

where:

- $\Phi_{\text{initial,C}}$ : Initial value of  $\Phi_C$  at  $s = 0$ .
- $\Phi_{\text{final,C}}$ : Final value of  $\Phi_C$  at  $s = 1$ .

The annealing parameter  $s \in [0, 1]$  parameterizes the evolution, and the bias  $c(s)$  is controlled by an external room-temperature current source, ensuring linear growth of the persistent current  $I_p(s)$  during the anneal. Consequently, the problem energy scale grows quadratically in time.

To map the QPU's Hamiltonian to the Ising Hamiltonian, the external flux  $\Phi_{xi}(s)$  and energy parameters  $A(s)$  and  $B(s)$  are set to:

$$\Phi_{xi}(s) = M_{AFM} |I_p(\Phi_C(s))|. \quad (3)$$

$$A(s) = \Delta_q(\Phi_C(s)) \quad (4)$$

$$B(s) = 2M_{AFM} |I_p(\Phi_C(s))|^2 \quad (5)$$

As  $\Phi_C(s)$  changes during the anneal,  $\Phi_{xi}(s)$  adjusts to maintain the relative energy ratio between the  $h_i$  and  $J_{ij}$  terms.

Thus, Eq. (1) can be rewritten as:

$$H(s) = -\frac{A(s)}{2} \sum_i \sigma_x^{(i)} + \frac{B(s)}{2} \left( \sum_i h_i \sigma_z^{(i)} + \sum_{i>j} J_{ij} \sigma_z^{(i)} \sigma_z^{(j)} \right) \quad (6)$$

At  $s = 0$ , the system is in the quantum ground state dominated by  $H_{\text{initial}}$  ( $A(0) \gg B(0)$ ). As  $s$  progresses,  $A(s)$  decreases while  $B(s)$  increases, transitioning the system into a classical regime. At  $s = 1$ , the final Hamiltonian describes a classical Ising spin model. Refer to the appendix for the physical properties of the Advantage™ 4.1 D-Wave QPU and Advantage™ 2.6 D-Wave QPU prototype and its data.

The quantum Hamiltonian  $H(s)$  is decomposed into two components:

- **Initial Hamiltonian  $H_{\text{initial}}$ :**

$$H_{\text{initial}} = \sum_i \sigma_x^{(i)}, \quad (7)$$

which governs quantum fluctuations.

- **Problem Hamiltonian  $H_p$ :**

$$H_{\text{Problem}} = \sum_i h_i \sigma_z^{(i)} + \sum_{i>j} J_{ij} \sigma_z^{(i)} \sigma_z^{(j)}, \quad (8)$$

which encodes the classical Ising model problem.

The Problem Hamiltonian  $H_p$  can be expressed in terms of the QUBO formulation. The equivalence is established by mapping the Ising variables  $\sigma_z^{(i)}$  to binary variables  $x_i \in \{0, 1\}$ :

$$H_p = \sum_i a_i x_i + \sum_{i>j} b_{ij} x_i x_j + c, \quad (9)$$

where the coefficients  $a_i$ ,  $b_{ij}$ , and  $c$  correspond to the Ising model parameters  $h_i$ ,  $J_{ij}$ , and a constant offset, respectively. This transformation enables leveraging QUBO solvers for optimization tasks.

For Solving the IK, it must first be converted into a QUBO formulation. This is then transformed into an Ising model. Afterward, by applying minor embedding, we embed the IK problem onto the Pegasus and Zephyr topologies. The Quantum Annealing process in the D-Wave QPU aims to find the ground state of  $H_p$ , corresponding to the optimal solution of the optimization problem. By starting with a known ground state of  $H_{\text{initial}}$  and evolving the system adiabatically through  $H(s)$ , the process converges to the ground state of  $H_p$ , thereby solving the problem efficiently. The entire pipeline, as depicted in the Figure 1, is followed to decode the results into the required joint angles.

### III. INVERSE KINEMATICS AS A QUBO

#### A. Linear Binary Approximation

We introduce a binary selector vector  $\mathbf{q} \in \mathbb{B}^m$ , for which one and only one of its elements  $q_i \in \{0, 1\}$  is equal to 1:

$$\sum_{i=1}^m q_i = 1. \quad (10)$$

Given a sequence of angles  $0 \geq \varphi_1 > \varphi_2 > \dots > \varphi_m \leq 2\pi$  we can find associated values of the trigonometric functions  $\cos(\varphi)$  and  $\sin(\varphi)$  and arrange them as elements of a vector:

$$\mathbf{t}_1 = [c_1 \quad \dots \quad c_n]^T, \quad c_i = \cos(\varphi_i), \quad (11)$$

$$\mathbf{t}_2 = [s_1 \quad \dots \quad s_n]^T, \quad s_i = \sin(\varphi_i); \quad (12)$$

we call  $\mathbf{t}_1, \mathbf{t}_2$  value vectors. With that, we can introduce a linear binary approximation (LBA) of the cosine and sine functions:

$$\cos(\varphi) \approx \mathbf{t}_1^T \mathbf{q}, \quad \sin(\varphi) \approx \mathbf{t}_2^T \mathbf{q}. \quad (13)$$

LBA is exact for the angles  $\varphi_i$  that were used to compute elements of the value vectors  $\mathbf{t}_1, \mathbf{t}_2$ . For any selector vector  $\mathbf{q}$  the fundamental identity holds:

$$(\mathbf{t}_1^T \mathbf{q})^2 + (\mathbf{t}_2^T \mathbf{q})^2 = 1. \quad (14)$$

#### B. Inverse Kinematics of a Planar Serial Linkage

Consider a planar kinematic chain with  $n$  links connected via rotary joints. Let the length of the  $j$ -th link be  $l_j$  and its orientation in the global frame be defined by the angle  $\varphi_j$ . Then the Cartesian coordinates  $(r_1, r_2)$  of the end effector placed at the end of the  $n$ -th link are given by the following expression:

$$r_1 = \sum_{j=1}^n l_j \cos(\varphi_j), \quad r_2 = \sum_{j=1}^n l_j \sin(\varphi_j). \quad (15)$$

To find angles  $\varphi_j^*$  corresponding to the desired position  $r_1^*, r_2^*$  of the end-effector we solve the following optimization problem:

$$\varphi_j = \underset{\varphi_j}{\operatorname{argmin}} (r_1^* - \sum_{j=1}^n l_j \cos(\varphi_j))^2 + (r_2^* - \sum_{j=1}^n l_j \sin(\varphi_j))^2$$

This represents an inverse kinematics (IK) problem cast as a nonconvex optimization with continuous variables. In

the next subsection, we approximate it as a quadratic binary optimization problem.

#### C. Inverse Kinematics as a QUBO

To approximate the IK problem described in the last subsection, we use  $n$  linear binary approximations  $\cos(\varphi_j) \approx \mathbf{t}_1^T \mathbf{q}_j$  and  $\sin(\varphi_j) \approx \mathbf{t}_2^T \mathbf{q}_j$ , each associated with its selector vector  $\mathbf{q}_j \in \mathbb{B}^m$ . Concatenation of all selector vectors is denoted as  $\bar{\mathbf{q}} = [\mathbf{q}_1^T, \dots, \mathbf{q}_n^T]^T$ .

For convenience, we define vector  $\mathbf{d} = [l_1, \dots, l_n]^T$ . The resulting optimization problem takes the form:

$$\underset{\bar{\mathbf{q}}}{\operatorname{minimize}} \sum_{k=1}^2 \left( \bar{\mathbf{q}}^T ((\mathbf{d}\mathbf{d}^T) \otimes (\mathbf{t}_k \mathbf{t}_k^T)) \bar{\mathbf{q}} - 2r_k (\mathbf{d} \otimes \mathbf{t}_k)^T \bar{\mathbf{q}} \right)$$

where  $\otimes$  is a Kronecker product.

To ensure that every vector  $\mathbf{q}_j$  abides by the selector constraints (10) we introduce a big-N penalty function  $p(\mathbf{q}_j)$ :

$$p(\mathbf{q}_j) = N(\mathbf{1}_m^T \mathbf{q}_j - 1)^2 \quad (16)$$

where  $\mathbf{1}_m \in \mathbb{R}^m$  is a vector of ones and  $N \gg 0$  is a sufficiently big number. With big-N penalty, the optimization takes the form:

$$\underset{\bar{\mathbf{q}}}{\operatorname{minimize}} \sum_{k=1}^2 \left( \bar{\mathbf{q}}^T ((\mathbf{d}\mathbf{d}^T) \otimes (\mathbf{t}_k \mathbf{t}_k^T)) \bar{\mathbf{q}} - 2r_k (\mathbf{d} \otimes \mathbf{t}_k)^T \bar{\mathbf{q}} \right) + N \left( \bar{\mathbf{q}}^T (\mathbf{I} \otimes (\mathbf{1}_m \mathbf{1}_m^T)) \bar{\mathbf{q}} - 2(\mathbf{1}_m \otimes \mathbf{1}_m)^T \bar{\mathbf{q}} \right) \quad (17)$$

Note that the constant components of the cost function are omitted, as they do not influence the optimal choice of the decision variable, denoted as  $\bar{\mathbf{q}}$ . The resulting problem is formulated as a Quadratic Unconstrained Binary Optimization (QUBO), which is compatible with quantum annealer hardware, such as the D-Wave Advantage. This problem will be embedded into a specific topology for an Ising model, where the variables in the QUBO are mapped (i.e. minor embedded [20]) to physical qubits, as illustrated in Fig. 3.

#### D. Higher-Order Inverse Kinematics Problems

Only a subset of inverse kinematics problems resolves to a QUBO form under the proposed linear binary approximation. Instead, the resulting problem will take a form of a higher-order unconstrained optimization (HUBO), unless a different approximation is used. HUBO problems cannot be solved by the quantum annealer hardware directly, but they can be approximated by a larger (in terms of the number of variables) QUBO problems sharing the same global minimum [21], [22], [23], [24], [25].

### IV. EMPIRICAL RESULTS AND ANALYSIS FOR SOLVING IK USING QUANTUM ANNEALING

To evaluate the performance of solving IK problem reformulated into a QUBO model using quantum annealing, two embedding strategies—**Global Embedding** and **Clique Embedding** [26]—were systematically analyzed. These methods were evaluated for their effectiveness in mapping the problem onto quantum annealing hardware. Specifically, the

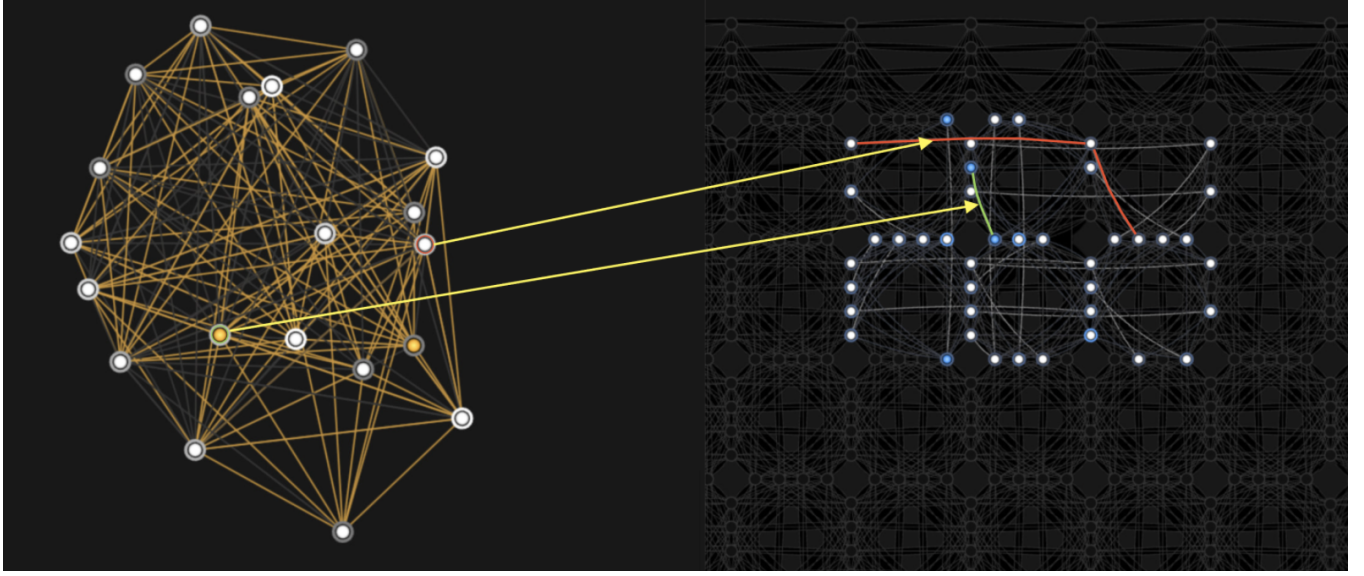


Fig. 3: Mapping a Two-Linked IK problem graph onto a quantum annealer’s hardware architecture—highlighting the embedding of a densely connected problem graph (left) onto the physical qubit connectivity of a D-Wave quantum processing unit (right). The yellow arrows illustrate the correspondence between logical nodes and their mapped qubits, while the red and green edges represent couplings between qubits in the hardware graph. The white nodes on the left figure and the right figure represent zeroes in the QUBO and Ising formulations, respectively, while the yellow and blue nodes represent ones in the QUBO and Ising formulations, respectively.

analysis focused on physical qubit utilization and quantum processing unit access times, leveraging two state-of-the-art hardware topologies: ”Pegasus [27] and ”Zephyr [28]”. The IK problem addressed was a Two-Linked Inverse Kinematics problem. Problem sizes ranged from  $N = 5$  to  $N = 10$ , corresponding to 10 to 20 logical qubits. Each instance of the QUBO problem required a matrix size of  $2N \times 2N$ , necessitating  $2N$  logical qubits to accurately represent the problem.

- **Global Embedding:** A general-purpose embedding tool which models all constraints collectively and maps the aggregate onto the QPU graph. This approach typically uses fewer qubits and shorter chains, making it more efficient for smaller problems. However, it may not be scalable for larger or more complex problems.
- **Clique Embedding:** A specialized embedding technique optimized for fully connected QUBOs.

The IK problem is not fully connected but exhibits dense connectivity, with the number of connections approximated as  $\frac{3}{2}N$ .

Two quantum annealing hardware topologies were considered:

- **Pegasus Topology** (Advantage System 4.1): Capable of embedding fully connected QUBOs with up to 177 logical qubits.
- **Zephyr Topology** (Advantage Prototype 2.6): Supports embedding fully connected QUBOs with up to 80 logical qubits, featuring higher local connectivity compared to Pegasus.

| Logical Qubits | $N_{GP}$  | $N_{CP}$ | $N_{GZ}$  | $N_{CZ}$ |
|----------------|-----------|----------|-----------|----------|
| 10             | 15        | 20       | <b>12</b> | 30       |
| 12             | <b>20</b> | 36       | <b>20</b> | 36       |
| 14             | 27        | 42       | <b>24</b> | 42       |
| 16             | 35        | 48       | <b>31</b> | 48       |
| 18             | 41        | 54       | <b>35</b> | 54       |
| 20             | 50        | 60       | <b>42</b> | 60       |

TABLE I: Average Physical Qubits for Different Embedding Techniques in Pegasus and Zephyr

#### A. Qubit Usage Analysis

The analysis began with the number of physical qubits required to represent logical variables under different embedding techniques and topologies. Table I presents the average physical qubit usage for various problem sizes and embedding strategies. The columns represent:

- $N_{GP}$ : Global Embedding on Pegasus
- $N_{CP}$ : Clique Embedding on Pegasus
- $N_{GZ}$ : Global Embedding on Zephyr
- $N_{CZ}$ : Clique Embedding on Zephyr

We observe that  $N_{GZ}$  is the most efficient in terms of physical qubit usage. The results are the average of multiple tests conducted for each embedding technique and topology.

#### B. QPU Access Time Analysis

In Table II, we present the average QPU access times for each embedding strategy. The columns represent:

- $t_{GP}$ : Global Embedding on Pegasus
- $t_{CP}$ : Clique Embedding on Pegasus
- $t_{GZ}$ : Global Embedding on Zephyr

- $t_{CZ}$ : Clique Embedding on Zephyr

| Logical Qubits | $t_{GP}$ (ms) | $t_{CP}$ (ms) | $t_{GZ}$ (ms) | $t_{CZ}$ (ms) |
|----------------|---------------|---------------|---------------|---------------|
| 10             | 584.4         | 648.76        | <b>413.3</b>  | 534.5         |
| 12             | 602.2         | 878.16        | <b>415.2</b>  | 542.1         |
| 14             | 611.4         | 878.16        | <b>490.1</b>  | 552.5         |
| 16             | 635.76        | 889.56        | 566.5         | <b>552.5</b>  |
| 18             | 618.3         | 1013.31       | <b>492.9</b>  | 560.1         |
| 20             | 540.0         | 1016.7        | <b>413.8</b>  | 567.7         |

TABLE II: Average QPU Access Time for Different Embedding Techniques in Pegasus and Zephyr (ms)

The results show that  $t_{GZ}$  was the best-performing configuration, aligning with the findings from the qubit usage analysis. Global Embedding is better suited for this problem because the QUBO is not fully connected. Global Embedding required fewer physical qubits compared to Clique Embedding, particularly for larger problem sizes. For smaller problem sizes, the dense local connectivity of the Zephyr topology provided a slight advantage in embedding efficiency. Clique Embedding exhibited longer QPU access times due to its general-purpose nature, whereas Global Embedding achieved faster programming but required manual optimization.

### C. Large-Scale QUBOs

For large-scale problem instances that exceed the limitations of purely quantum solvers, we utilize the Hybrid D-Wave Solver. This hybrid solver integrates quantum annealing with classical optimization techniques to efficiently solve large-scale QUBO problems as shown in Figure 4.

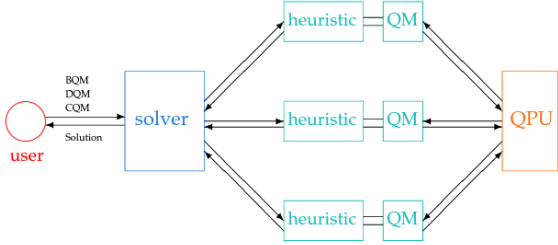


Fig. 4: Structure of a hybrid solver in D-Wave’s hybrid solver service. Adapted from [29]

In Figure 5, the difference in time-to-solution (TTS) among Quantum Annealing (QA), Simulated Annealing (SA) [30], [31], and the Probabilistic Tensor Train Sampler (PROTES), which employs probabilistic sampling from a probability density function represented in a low-parametric tensor train format [32], is illustrated <sup>1</sup>. For a QUBO instance of size  $900 \times 900$  (corresponding to  $2^{900}$  possible solutions), QA demonstrates a speedup of more than 30 times compared to classical methods. Although all three methods produce solutions of comparable quality, simulated annealing

<sup>1</sup>In these experiments, number of executed samples in SA and QA were 5000, while the PROTES number of allowed requests to the objective function—was set to 2000

marginally outperforms quantum annealing in a few instances due to the Integrated Control Errors (ICE) effect. Specifically, the dynamic range of  $h$  and  $J$  values may be limited by integrated control errors (ICE). Instead of finding low-energy states for an optimization problem defined by  $h$  and  $J$  as in equation 1, the quantum processing unit solves a slightly altered problem that can be modeled as:

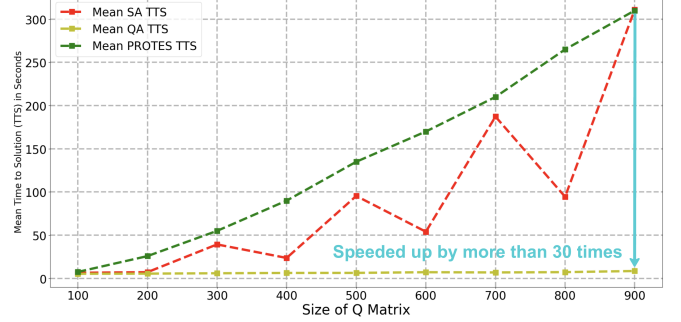


Fig. 5: Mean TTS for QA, SA, and PROTES on large-scale Two-Linked IK QUBO instances.

$$E_{\text{disj}}(s) = \sum_{i=1}^N (h_i + \delta h_i) s_i + \sum_{i=1}^N \sum_{j=i+1}^N (J_{i,j} + \delta J_{i,j}) s_i s_j, \quad (18)$$

where  $\delta h_i$  and  $\delta J_{i,j}$  characterize the errors in the parameters  $h_i$  and  $J_{i,j}$ , respectively [33]. Because  $\delta h$  and  $\delta J$  are summed over  $N$ , fidelity limitations tend to have a greater effect on performance for full-QPU-sized problems, for a given dynamic range and distribution of  $h$  and  $J$ . This can result in slightly different solutions compared to the ideal case.

The results in Figure 5 emphasize the significant advantage of the Hybrid D-Wave Solver in reducing time-to-solution for practical, large-scale optimization problems. The hybrid approach effectively combines the strengths of quantum and classical techniques, making it a robust solution for tackling computationally intensive QUBO instances. However, its performance is limited by the overhead and variability introduced by integrating these two components, which means that for certain problems it may not achieve the optimal solution. Reliance on classical processing can restrict the overall speedup and may prevent consistently achieving the global optimum.

## V. CONCLUSIONS

The application of quantum annealing to IK problem, reformulated as a QUBO, demonstrates the potential of quantum computing to address complex optimization challenges in robotics. Among the embedding strategies analyzed, Global Embedding on Zephyr proved to be the most efficient, offering lower physical qubit usage and faster QPU access times. These results highlight the advantages of hybrid quantum-classical approaches, achieving more than a 30-fold speedup in optimization tasks. Future work will focus on

optimizing embedding strategies and utilizing different quantum annealing protocols to further enhance the performance of the solution.

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## APPENDIX

### PHYSICAL PROPERTIES DATA OF THE ADVANTAGE™ 4.1 D-WAVE QPU AND ADVANTAGE™ 2.6 D-WAVE QPU PROTOTYPE

The Quantum Annealing processes in the Advantage™ 4.1 D-Wave QPU and the Advantage™ 2.6 D-Wave QPU prototype differ. As seen in Equation 1, the annealing process depends on various parameters, the values of which are summarized in Table III. Consequently, the quantum annealing schedule differs, as illustrated in Figures 6 and 7.

| Parameter                                                                                                                                                      | Advantage™ 4.1                 | Advantage™ 2 Prototype |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|------------------------|
| Model                                                                                                                                                          | Advantage™, performance update | Advantage™ 2 Prototype |
| Graph size                                                                                                                                                     | P16                            | Z6                     |
| Qubits                                                                                                                                                         | 5627                           | 1215                   |
| Couplers                                                                                                                                                       | 40279                          | 10788                  |
| Qubit temperature (mK)                                                                                                                                         | $15.4 \pm 0.1$                 | $16.5 \pm 1.0$         |
| Maximum available mutual inductance (MAFM, pH)                                                                                                                 | 1.647                          | 0.443                  |
| Quantum critical point (GHz)                                                                                                                                   | 1.391                          | 2.014                  |
| Qubit inductance ( $L_q$ , pH)                                                                                                                                 | 371.683                        | 106.617                |
| Qubit capacitance ( $C_q$ , fF)                                                                                                                                | 118.606                        | 207.188                |
| Qubit critical current ( $I_c$ , $\mu A$ )                                                                                                                     | 2.099                          | 4.570                  |
| Average single qubit thermal width (Ising units)                                                                                                               | 0.198                          | 0.117                  |
| FM problem freezeout (scaled time)                                                                                                                             | 0.064                          | 0.009                  |
| Single qubit freezeout (scaled time)                                                                                                                           | 0.612                          | 0.605                  |
| Initial flux for CCJJ structures ( $\Phi_{i,CCJJ}$ ( $\Phi_0$ ))                                                                                               | -0.621                         | -0.726                 |
| Final flux for CCJJ structures ( $\Phi_{f,CCJJ}$ ( $\Phi_0$ ))                                                                                                 | -0.717                         | -0.819                 |
| Readout time range ( $\mu s$ ) (Typical readout times for reading between one qubit and the full QPU.)                                                         | 17.0 to 235.0                  | 17.0 to 87.0           |
| Programming time ( $\mu s$ ) (Typical for problems run on this QPU. Actual problem programming times may vary slightly depending on the nature of the problem) | $\sim 14100$                   | $\sim 18200$           |
| QPU delay time per sample ( $\mu s$ )                                                                                                                          | 20.5                           | 20.5                   |
| Readout error rate (Error rate when reading the full system)                                                                                                   | $\leq 0.001$                   | $\leq 0.001$           |

TABLE III: Physical Properties of the Advantage™ 4.1 and Advantage™ 2 Prototype D-Wave QPUs.

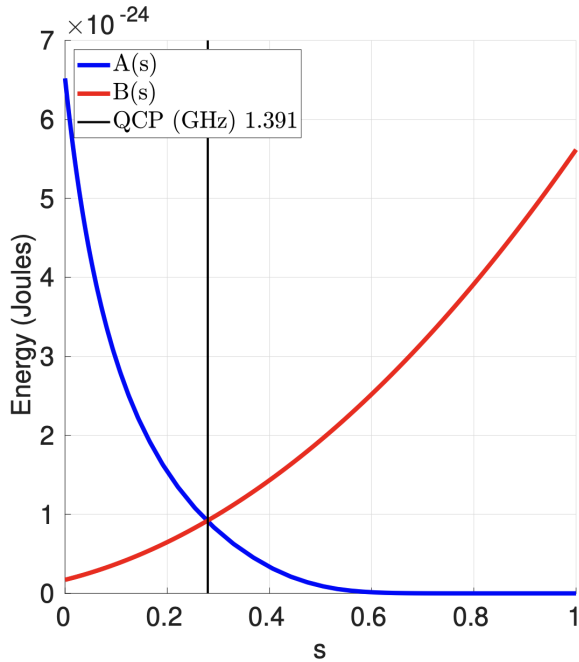


Fig. 6: Standard annealing schedule for the QPU Advantage 4.1, showing energy changes as a function of scaled time. Adapted from [18]

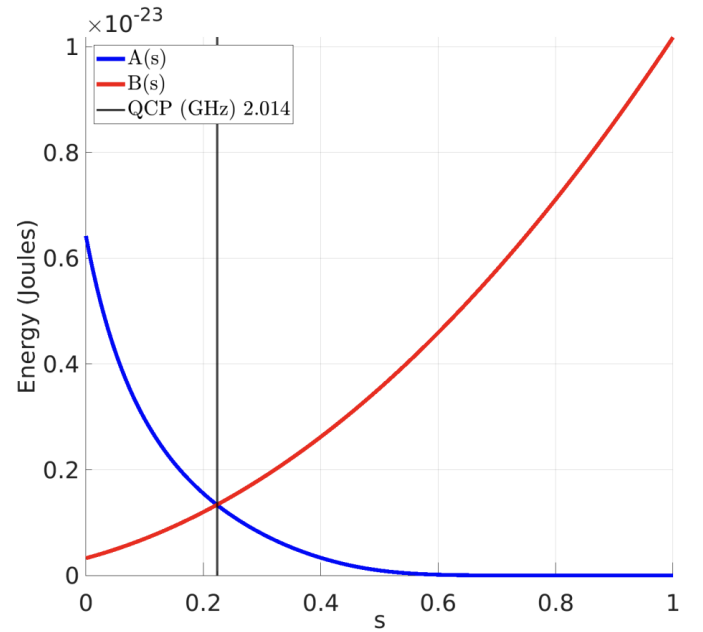


Fig. 7: Standard annealing schedule for the QPU Advantage Prototype 2.6, showing energy changes as a function of scaled time. Adapted from [18]